Etapa 1 – importação

Ambiente: Ubuntu Server, com MongoDB executado no contêiner `mongo-atividade1-v7`.

O arquivo `restaurantes.json` foi copiado para o contêiner e importado para o banco

`atividade1`, na coleção `restaurants`.

docker cp restaurantes.json mongo-atividade1-v7:/tmp/restaurantes.json

docker exec mongo-atividade1-v7 mongoimport --db atividade1 --collection restaurants --file /tmp/restaurantes.json

Resultado: 25.359 documentos importados com sucesso e 0 falhas.

```
davi@davi:~/nosql$ docker exec mongo-atividade1-v7 mongosh --quiet --eval 'db.getSiblingDB("atividade1").restaurants.countDocuments({})'
25359
davi@davi:~/nosql$
```

Etapa 2 – inserção

- docker exec -it mongo-atividade1-v7 mongosh -> shell do mongo

```
already on db atividade1
atividade1> db.restaurants.insertOne({
|   address: {
|     building: "100",
|     coord: [-73.9700, 40.7800],
|     street: "Example Street",
|     zipcode: "10022"
|   },
|   borough: "Manhattan",
|   cuisine: "Italian",
|   grades: [
|     { grade: "A", score: 8 }
|   ],
|   name: "Restaurante Exemplo Atividade 1",
|   restaurant_id: "atividade1-manual-001"
| })
|
| db.restaurants.findOne({ restaurant_id: "atividade1-manual-001" })
{
  _id: ObjectId('6ab7db621d29061eb6dbf93d'),
  address: {
    building: '100',
    coord: [ -73.97, 40.78 ],
    street: 'Example Street',
    zipcode: '10022'
  },
  borough: 'Manhattan',
  cuisine: 'Italian',
  grades: [ { grade: 'A', score: 8 } ],
  name: 'Restaurante Exemplo Atividade 1',
  restaurant_id: 'atividade1-manual-001'
}
atividade1>
```

ETAPA 3 – CONSULTAS

1. Listar todos os restaurantes: db.restaurants.find({})

```
test> db.restaurants.find({})

test> use atividade1
switched to db atividade1
atividade1> db.restaurants.find({})
[
  {
    _id: ObjectId('6ab7d9318ef433a6060d9e7d'),
    address: {
      building: '1007',
      coord: [ -73.856077, 40.848447 ],
      street: 'Morris Park Ave',
      zipcode: '10462'
    },
    borough: 'Bronx',
    cuisine: 'Bakery',
    grades: [
      { date: ISODate('2014-03-03T00:00:00.000Z'), grade: 'A', score: 2 },
      { date: ISODate('2013-09-11T00:00:00.000Z'), grade: 'A', score: 6 },
      { date: ISODate('2013-01-24T00:00:00.000Z'), grade: 'A', score: 10 },
      { date: ISODate('2011-11-23T00:00:00.000Z'), grade: 'A', score: 9 },
      { date: ISODate('2011-03-10T00:00:00.000Z'), grade: 'B', score: 14 }
    ],
    name: 'Morris Park Bake Shop',
    restaurant_id: '30075445'
  },
  {
    _id: ObjectId('6ab7d9318ef433a6060d9e7e'),
    address: {
      building: '469',
      coord: [ -73.961704, 40.662942 ],
      street: 'Flatbush Avenue',
      zipcode: '11225'
    },
    borough: 'Brooklyn',
    cuisine: 'Hamburgers',
    grades: [
      { date: ISODate('2014-12-30T00:00:00.000Z'), grade: 'A', score: 8 },
```

2. Restaurantes de Manhattan: db.restaurants.find({ borough: "Manhattan" })

```
atividade1> db.restaurants.find({ borough: "Manhattan" })
[
  {
    _id: ObjectId('6ab7d9318ef433a6060d9e7f'),
    address: {
      building: '351',
      coord: [ -73.98513559999999, 40.7676919 ],
      street: 'West   57 Street',
      zipcode: '10019'
    },
    borough: 'Manhattan',
    cuisine: 'Irish',
    grades: [
      { date: ISODate('2014-09-06T00:00:00.000Z'), grade: 'A', score: 2 },
      { date: ISODate('2013-07-22T00:00:00.000Z'), grade: 'A', score: 11 },
      { date: ISODate('2012-07-31T00:00:00.000Z'), grade: 'A', score: 12 },
      { date: ISODate('2011-12-29T00:00:00.000Z'), grade: 'A', score: 12 }
    ],
    name: 'Dj Reynolds Pub And Restaurant',
    restaurant_id: '30191841'
  },
  {
    _id: ObjectId('6ab7d9318ef433a6060d9e8a'),
    address: {
      building: '1',
      coord: [ -73.96926909999999, 40.7685235 ],
      street: 'East   66 Street',
      zipcode: '10065'
    },
    borough: 'Manhattan',
    cuisine: 'American',
    grades: [
      { date: ISODate('2014-05-07T00:00:00.000Z'), grade: 'A', score: 3 },
      { date: ISODate('2013-05-03T00:00:00.000Z'), grade: 'A', score: 4 },
```

3. Restaurantes do CEP 10022: db.restaurants.find({ "address.zipcode": "10022" })

```
atividade1> db.restaurants.find({ "address.zipcode": "10022" })
[
  {
    _id: ObjectId('6ab7d9318ef433a6060d9ea8'),
    address: {
      building: '437',
      coord: [ -73.975393, 40.757365 ],
      street: 'Madison Avenue',
      zipcode: '10022'
    },
    borough: 'Manhattan',
    cuisine: 'American',
    grades: [
      { date: ISODate('2014-06-03T00:00:00.000Z'), grade: 'A', score: 9 },
      { date: ISODate('2013-06-07T00:00:00.000Z'), grade: 'A', score: 5 },
      { date: ISODate('2012-06-29T00:00:00.000Z'), grade: 'A', score: 12 },
      { date: ISODate('2012-02-06T00:00:00.000Z'), grade: 'A', score: 11 },
      { date: ISODate('2011-06-23T00:00:00.000Z'), grade: 'A', score: 13 }
    ],
    name: 'Berkely',
    restaurant_id: '40363685'
  },
  {
    _id: ObjectId('6ab7d9318ef433a6060d9eb7'),
    address: {
      building: '1',
      coord: [ -73.97166039999999, 40.764832 ],
      street: 'East   60 Street',
      zipcode: '10022'
    },
    borough: 'Manhattan',
    cuisine: 'American',
    grades: [
      { date: ISODate('2014-10-16T00:00:00.000Z'), grade: 'B', score: 24 },
      { date: ISODate('2014-05-02T00:00:00.000Z'), grade: 'A', score: 4 },
```

4. Restaurantes que possuam alguma nota B: db.restaurants.find({ "grades.grade": "B" })

```
atividade1> db.restaurants.find({ "grades.grade": "B" })
[
  {
    _id: ObjectId('6ab7d9318ef433a6060d9e7d'),
    address: {
      building: '1007',
      coord: [ -73.856077, 40.848447 ],
      street: 'Morris Park Ave',
      zipcode: '10462'
    },
    borough: 'Bronx',
    cuisine: 'Bakery',
    grades: [
      { date: ISODate('2014-03-03T00:00:00.000Z'), grade: 'A', score: 2 },
      { date: ISODate('2013-09-11T00:00:00.000Z'), grade: 'A', score: 6 },
      { date: ISODate('2013-01-24T00:00:00.000Z'), grade: 'A', score: 10 },
      { date: ISODate('2011-11-23T00:00:00.000Z'), grade: 'A', score: 9 },
      { date: ISODate('2011-03-10T00:00:00.000Z'), grade: 'B', score: 14 }
    ],
    name: 'Morris Park Bake Shop',
    restaurant_id: '30075445'
  },
  {
    _id: ObjectId('6ab7d9318ef433a6060d9e7e'),
    address: {
      building: '469',
      coord: [ -73.961704, 40.662942 ],
      street: 'Flatbush Avenue',
      zipcode: '11225'
    },
    borough: 'Brooklyn',
    cuisine: 'Hamburgers',
    grades: [
      { date: ISODate('2014-12-30T00:00:00.000Z'), grade: 'A', score: 8 },
      { date: ISODate('2014-07-01T00:00:00.000Z'), grade: 'B', score: 23 },
      { date: ISODate('2013-04-30T00:00:00.000Z'), grade: 'A', score: 12 },
      { date: ISODate('2012-05-08T00:00:00.000Z'), grade: 'A', score: 12 }
```

ETAPA 4 – OPERADORES

1. Restaurantes com alguma pontuação maior que 30: db.restaurants.find({ "grades.score": { $gt: 30 } })

```
atividade1>

atividade1> db.restaurants.find({ "grades.score": { $gt: 30 } })
[
  {
    _id: ObjectId('6ab7d9318ef433a6060d9e82'),
    address: {
      building: '8825',
      coord: [ -73.8803827, 40.7643124 ],
      street: 'Astoria Boulevard',
      zipcode: '11369'
    },
    borough: 'Queens',
    cuisine: 'American',
    grades: [
      { date: ISODate('2014-11-15T00:00:00.000Z'), grade: 'Z', score: 38 },
      { date: ISODate('2014-05-02T00:00:00.000Z'), grade: 'A', score: 10 },
      { date: ISODate('2013-03-02T00:00:00.000Z'), grade: 'A', score: 7 },
      { date: ISODate('2012-02-10T00:00:00.000Z'), grade: 'A', score: 13 }
    ],
    name: 'Brunos On The Boulevard',
    restaurant_id: '40356151'
  },
  {
    _id: ObjectId('6ab7d9318ef433a6060d9e89'),
    address: {
      building: '1269',
      coord: [ -73.871194, 40.6730975 ],
      street: 'Sutter Avenue',
      zipcode: '11208'
    },
    borough: 'Brooklyn',
    cuisine: 'Chinese',
    grades: [
      { date: ISODate('2014-09-16T00:00:00.000Z'), grade: 'B', score: 21 },
      { date: ISODate('2013-08-28T00:00:00.000Z'), grade: 'A', score: 7 },
      { date: ISODate('2013-04-02T00:00:00.000Z'), grade: 'C', score: 56 },
      { date: ISODate('2012-08-15T00:00:00.000Z'), grade: 'B', score: 27 },
      { date: ISODate('2012-03-28T00:00:00.000Z'), grade: 'B', score: 27 }
    ],
    name: 'May May Kitchen',
    restaurant_id: '40358429'
  },
  {
    _id: ObjectId('6ab7d9318ef433a6060d9eaf'),
    address: {
      building: '0',
      coord: [ -84.2040813, 9.9986585 ],
      street: 'Guardia Airport Parking',
      zipcode: '11371'
    },
    borough: 'Queens',
    cuisine: 'American',
    grades: [
      { date: ISODate('2014-05-16T00:00:00.000Z'), grade: 'A', score: 9 },
      { date: ISODate('2013-05-10T00:00:00.000Z'), grade: 'A', score: 10 },
      { date: ISODate('2012-05-15T00:00:00.000Z'), grade: 'A', score: 9 },
      { date: ISODate('2011-11-02T00:00:00.000Z'), grade: 'C', score: 32 }
    ],
    name: 'Terminal Cafe/Yankee Clipper',
    restaurant_id: '40364262'
```

2. Restaurantes com alguma pontuação menor que 10: db.restaurants.find({ "grades.score": { $lt: 10 } })

```
atividade1>

atividade1> db.restaurants.find({ "grades.score": { $lt: 10 } })
[
  {
    _id: ObjectId('6ab7d9318ef433a6060d9e7d'),
    address: {
      building: '1007',
      coord: [ -73.856077, 40.848447 ],
      street: 'Morris Park Ave',
      zipcode: '10462'
    },
    borough: 'Bronx',
    cuisine: 'Bakery',
    grades: [
      { date: ISODate('2014-03-03T00:00:00.000Z'), grade: 'A', score: 2 },
      { date: ISODate('2013-09-11T00:00:00.000Z'), grade: 'A', score: 6 },
      { date: ISODate('2013-01-24T00:00:00.000Z'), grade: 'A', score: 10 },
      { date: ISODate('2011-11-23T00:00:00.000Z'), grade: 'A', score: 9 },
      { date: ISODate('2011-03-10T00:00:00.000Z'), grade: 'B', score: 14 }
    ],
    name: 'Morris Park Bake Shop',
    restaurant_id: '30075445'
  },
  {
    _id: ObjectId('6ab7d9318ef433a6060d9e7e'),
    address: {
      building: '469',
      coord: [ -73.961704, 40.662942 ],
      street: 'Flatbush Avenue',
      zipcode: '11225'
    },
    borough: 'Brooklyn',
    cuisine: 'Hamburgers',
    grades: [
      { date: ISODate('2014-12-30T00:00:00.000Z'), grade: 'A', score: 8 },
      { date: ISODate('2014-07-01T00:00:00.000Z'), grade: 'B', score: 23 },
      { date: ISODate('2013-04-30T00:00:00.000Z'), grade: 'A', score: 12 },
      { date: ISODate('2012-05-08T00:00:00.000Z'), grade: 'A', score: 12 }
    ],
    name: "Wendy'S",
    restaurant_id: '30112340'
  },
  {
    _id: ObjectId('6ab7d9318ef433a6060d9e7f'),
    address: {
      building: '351',
      coord: [ -73.98513559999999, 40.7676919 ],
      street: 'West   57 Street',
      zipcode: '10019'
    },
    borough: 'Manhattan',
    cuisine: 'Irish',
    grades: [
      { date: ISODate('2014-09-06T00:00:00.000Z'), grade: 'A', score: 2 },
      { date: ISODate('2013-07-22T00:00:00.000Z'), grade: 'A', score: 11 },
      { date: ISODate('2012-07-31T00:00:00.000Z'), grade: 'A', score: 12 },
      { date: ISODate('2011-12-29T00:00:00.000Z'), grade: 'A', score: 12 }
```

3. Culinária italiana E CEP 10075: db.restaurants.find({ cuisine: "Italian", "address.zipcode": "10075" })

```
atividade1>

atividade1> db.restaurants.find({ cuisine: "Italian", "address.zipcode": "10075" })
|
[
  {
    _id: ObjectId('6ab7d9318ef433a6060d9fd5'),
    address: {
      building: '1133',
      coord: [ -73.958885, 40.7745559 ],
      street: 'Lexington Avenue',
      zipcode: '10075'
    },
    borough: 'Manhattan',
    cuisine: 'Italian',
    grades: [
      { date: ISODate('2014-08-11T00:00:00.000Z'), grade: 'A', score: 11 },
      { date: ISODate('2013-12-10T00:00:00.000Z'), grade: 'A', score: 9 },
      { date: ISODate('2013-06-10T00:00:00.000Z'), grade: 'A', score: 12 },
      { date: ISODate('2012-06-08T00:00:00.000Z'), grade: 'A', score: 13 },
      { date: ISODate('2012-01-25T00:00:00.000Z'), grade: 'A', score: 8 },
      { date: ISODate('2011-09-13T00:00:00.000Z'), grade: 'A', score: 12 }
    ],
    name: 'Don Filippo Restaurant',
    restaurant_id: '40372417'
  },
  {
    _id: ObjectId('6ab7d9318ef433a6060da025'),
    address: {
      building: '1494',
      coord: [ -73.955387, 40.772368 ],
      street: 'Second Avenue',
      zipcode: '10075'
    },
    borough: 'Manhattan',
    cuisine: 'Italian',
    grades: [
      { date: ISODate('2014-01-23T00:00:00.000Z'), grade: 'A', score: 7 },
      { date: ISODate('2013-07-11T00:00:00.000Z'), grade: 'A', score: 11 },
      { date: ISODate('2013-01-22T00:00:00.000Z'), grade: 'A', score: 7 },
      { date: ISODate('2012-05-08T00:00:00.000Z'), grade: 'A', score: 10 },
      { date: ISODate('2011-11-02T00:00:00.000Z'), grade: 'A', score: 11 }
    ],
    name: "Lusardi'S Restaurant",
    restaurant_id: '40376142'
  },
  {
    _id: ObjectId('6ab7d9318ef433a6060da102'),
    address: {
      building: '1396',
      coord: [ -73.9573454, 40.7747147 ],
      street: 'Third Avenue',
      zipcode: '10075'
    },
    borough: 'Manhattan',
    cuisine: 'Italian',
    grades: [
      { date: ISODate('2014-02-20T00:00:00.000Z'), grade: 'A', score: 5 },
      { date: ISODate('2013-01-24T00:00:00.000Z'), grade: 'A', score: 13 },
      { date: ISODate('2011-12-27T00:00:00.000Z'), grade: 'A', score: 9 }
    ],
    name: 'Due',
    restaurant_id: '40386999'
```

4. Culinária italiana OU CEP 10075: db.restaurants.find({ $or: [{ cuisine: "Italian" }, { "address.zipcode": "10075" }] })

```
atividade1> db.restaurants.find({ $or: [ { cuisine: "Italian" }, { "address.zipcode": "10075" } ] })
|
[
  {
    _id: ObjectId('6ab7d9318ef433a6060d9eb5'),
    address: {
      building: '10004',
      coord: [ -74.03400479999999, 40.6127077 ],
      street: '4 Avenue',
      zipcode: '11209'
    },
    borough: 'Brooklyn',
    cuisine: 'Italian',
    grades: [
      { date: ISODate('2014-02-25T00:00:00.000Z'), grade: 'A', score: 12 },
      { date: ISODate('2013-06-27T00:00:00.000Z'), grade: 'A', score: 7 },
      { date: ISODate('2012-12-03T00:00:00.000Z'), grade: 'A', score: 10 },
      { date: ISODate('2011-11-09T00:00:00.000Z'), grade: 'A', score: 12 }
    ],
    name: 'Philadelhia Grille Express',
    restaurant_id: '40364305'
  },
  {
    _id: ObjectId('6ab7d9318ef433a6060d9eba'),
    address: {
      building: '1028',
      coord: [ -73.966032, 40.762832 ],
      street: '3 Avenue',
      zipcode: '10065'
    },
    borough: 'Manhattan',
    cuisine: 'Italian',
    grades: [
      { date: ISODate('2014-09-16T00:00:00.000Z'), grade: 'A', score: 13 },
      { date: ISODate('2014-02-24T00:00:00.000Z'), grade: 'A', score: 10 },
      { date: ISODate('2013-05-03T00:00:00.000Z'), grade: 'A', score: 10 },
      { date: ISODate('2012-08-20T00:00:00.000Z'), grade: 'A', score: 7 },
      { date: ISODate('2012-02-13T00:00:00.000Z'), grade: 'A', score: 9 }
    ],
    name: 'Isle Of Capri Resturant',
    restaurant_id: '40364373'
  },
  {
    _id: ObjectId('6ab7d9318ef433a6060d9ec8'),
    address: {
      building: '251',
      coord: [ -73.9775552, 40.7432016 ],
      street: 'East   31 Street',
      zipcode: '10016'
    },
    borough: 'Manhattan',
    cuisine: 'Italian',
    grades: [
      { date: ISODate('2014-04-22T00:00:00.000Z'), grade: 'A', score: 13 },
      { date: ISODate('2013-06-19T00:00:00.000Z'), grade: 'C', score: 32 },
      { date: ISODate('2012-05-22T00:00:00.000Z'), grade: 'A', score: 12 }
    ],
    name: 'Marchis Restaurant',
    restaurant_id: '40364668'
  },
  {
    _id: ObjectId('6ab7d9318ef433a6060d9ed4'),
    address: {
      building: '67',
```